A More Comprehensive Analog LDO Design and Simulation

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***Abstract*—This paper presents the design and simulation of a high-performance Low-Dropout (LDO) linear regulator implemented in a 180-nm CMOS process. To overcome the stability-bandwidth trade-offs inherent in driving large PMOS pass elements, the architecture incorporates an ultra-low-output-resistance dual-feedback buffer between the error amplifier and the power transistor. This structure pushes the non-dominant gate pole to higher frequencies, significantly enhancing loop stability and transient response without requiring large external capacitors. Operating from a supply range of 1.4 V to 2.5 V, the LDO provides a regulated 1.2 V output with a maximum load current of 40 mA With a design-optimized quiescent current of $100\,\mu$A, the regulator achieves a DC loop gain of 96 dB and a low-frequency PSRR of > 70 dB. Extensive temperature simulations across -40°C to 125°C demonstrate exceptional thermal stability, yielding an output voltage drift of only 0.52 mV and a calculated temperature coefficient of 2.62 ppm/$^\circ$C6.**

***Index Terms*—LDO, error amplifier, ultra-low output resistance buffer**

I. Introduction

Portable battery-powered systems and mixed-signal SoCs increasingly require low-noise, fast-response, and energy-efficient power regulation blocks. Low-dropout (LDO) linear regulators remain attractive in these environments due to their small area, low output noise and simple integration compared to switching regulators. However, as technology scales and output capacitors shrink for on-chip integration, conventional LDOs face significant stability challenges: the large gate capacitance of the pass element introduces non-dominant poles close to the unity-gain frequency, degrading phase margin, transient response, and PSRR. Traditional compensation schemes such as ESR-zero shaping or Miller nesting offer partial improvements but rely on large external components, consume higher quiescent current, and often fail at low supply voltage conditions.

Recent research has demonstrated that incorporating a low-impedance buffer between the error amplifier and the power transistor can effectively push non-dominant poles to higher frequencies, improving stability and transient settling while reducing dependency on external output capacitors. In particular, reference work on low-impedance dual-feedback buffers shows that shunt-feedback structures can achieve orders-of-magnitude reduction in buffer output resistance, enabling stable operation under wide load and voltage variations and significantly enhancing line/load regulation performance . Inspired by this direction, our project investigates an LDO architecture that embeds an ultra-low-output-resistance buffer to strengthen loop gain, improve dynamic response, and support low-dropout operation with minimal output capacitance.

The work presented in this report summarizes the design and implementation of a 180-nm CMOS LDO that targets low quiescent current, high stability margin, and fast transient behavior for integrated applications. Emphasis is placed on error amplifier development, output-impedance reduction through buffering, and frequency compensation optimization. Simulation results demonstrate improved output impedance, enhanced settling response, and competitive PSRR, validating the design approach and confirming the architectural advantages over traditional LDO structures.

II. LDO System Architecture

The block diagram in Fig. 1 shows the basic signal flow of our LDO. The reference voltage is set to 0.8 V, and the output is defined by the standard feedback relation

$$V_{\text{out}} = V_{\text{ref}}\left(1 + \frac{R_0}{R_1}\right)$$

where the resistive divider not only defines the output voltage but also feeds back real-time information from the load node. Any deviation in V_{out} directly appears at the EA input as an error signal. This implements the core principle of feedback regulation

$$V_{\text{error}} = V_{\text{ref}} - \frac{R_1}{R_0 + R_1} V_{\text{out}}$$

The error amplifier then translates this differential error into a high-gain control signal, enforcing closed-loop accuracy and DC load regulation.

However, unlike conventional LDO structures where the error amplifier must directly drive the large gate capacitance of the pass transistor, our architecture adds an ultra-low-output-impedance buffer between the amplifier and the pass device. In this structure, the buffer separates the precision control stage and the power driving stage, allowing the regulator to maintain high loop gain without compromising stability. By driving the PMOS gate through the buffer, the effective output resistance at the gate node is significantly reduced, which pushes the non-dominant pole to a higher frequency and improves overall dynamic performance.

The PMOS pass device regulates current conduction into the output node, closing the loop. The feedback divider senses the real output under load, translating voltage changes into proportional error-amplifier input variations. In steady state, the loop naturally drives the regulation error toward zero, forcing the output voltage to match the theoretical divider value and cancelling disturbances in supply or load.

III. Error Amplifier Design

The error amplifier is a key block in the LDO control loop because it determines the overall DC accuracy, loop gain and frequency behavior. In this design as shown in Fig. 2, we use a classical two-stage operational amplifier topology rather than a folded-cascode or telescopic structure. The main reason for this choice is the need for relatively high voltage gain while keeping the output swing wide enough to drive the buffer input. Telescopic or folded-cascode amplifiers could provide higher intrinsic gain, but they generally suffer from limited output swing and reduced gain when operating from low supply voltages. A single-stage topology was also considered, but its gain would not be sufficient to support tight regulation and strong PSRR. Therefore, the two-stage approach offers the right balance among gain, swing, and design complexity for this application.

The error amplifier is implemented as a two-stage voltage amplifier in order to provide enough loop gain and bandwidth for accurate regulation. The first stage is an NMOS differential pair with active PMOS load. It senses the reference voltage and the divided output voltage and converts their difference into a differential current. The second stage is a common-source gain stage that provides additional gain and large output swing to drive the following buffer. In small signal, the total DC gain can be written as the product of the two stage gains,

$$A_0 \approx A_1 \times A_2 \approx g_{m1} r_{o1} \cdot g_{m5} r_{o5}$$

where g_{m1}, r_{o1} are the effective transconductance and output resistance of the input differential pair, and g_{m5}, r_{o5} correspond to the main gain transistor in the second stage. In the design the second-stage transistor is made relatively wide with 24 fingers so that the $g_m r_o$ product is high without pushing the bias current too much. This gives a simulated DC gain of about 72 dB, which is sufficient to meet the static accuracy and PSRR requirements of the LDO.

To stabilize the two-stage structure and shape the loop dynamics, an RC compensation network is added between the first and second stages. The capacitor C_0 is connected in a Miller-like configuration and sets the dominant pole at the amplifier output,

$$p_{\text{dom}} \approx -\frac{1}{R_{\text{out,EA}} C_0}$$

where $R_{out,EA}$ is the small-signal output resistance of the second stage. The series resistor R_0 introduces a left-half-plane zero,

$$z_{\text{LHP}} \approx -\frac{1}{R_0 C_0}$$

which partially cancels the phase lag from the non-dominant pole and helps to increase the phase margin. By choosing $R_0 = 740\Omega$ and $C_0 = 330pF$, the dominant pole is placed well below the unity-gain frequency while the zero appears slightly before f_{UGF}. This arrangement results in a simulated unity-gain frequency of about 162 kHz and a phase margin close to 90° as shown in Fig. 3, indicating a robust and well-damped response over process and load variations.

The bias currents of both stages are selected as a compromise between noise, speed and quiescent power consumption. A higher current would push the unity-gain frequency further but increase the static current of the regulator, while too little current would limit bandwidth and slow transient response. With the chosen sizing and biasing, the error amplifier achieves high gain, moderate bandwidth and comfortable phase margin, providing a stable foundation for the feedback loop and enabling the buffer stage to focus primarily on driving the large pass-gate capacitance.

IV. Ultra-Low Output Resistance Buffer

Fig. 4 first shows two commonly used source follower structures. The conventional source follower in Fig. 4(a) only uses one transistor to drive the output node. Its output resistance is approximately

$$R_{\text{out,conv}} \approx \frac{1}{g_m}$$

so reducing the output impedance requires either increasing device size or bias current. This approach increases parasitics and power consumption, and also performs poorly at low supply voltage because the device may enter the linear region and lose transconductance. These limitations lead to weak drive capability, low bandwidth, and poor low-voltage behavior.

The super source follower in Fig. 4(b) improves this situation by adding a gain-boosting transistor and forming a local shunt-feedback loop. This increases the effective transconductance at the output node and lowers the resistance compared to the basic follower,

$$R_{\text{out,ssf}} \approx \frac{1}{g_m(1 + A_{\text{fb}})}$$

However, the improvement is still not sufficient for LDO use. As shown in the reference paper, when the supply voltage becomes low or the transistor moves close to threshold, the feedback gain decreases and the output resistance rises again. Therefore, even the super source follower cannot maintain

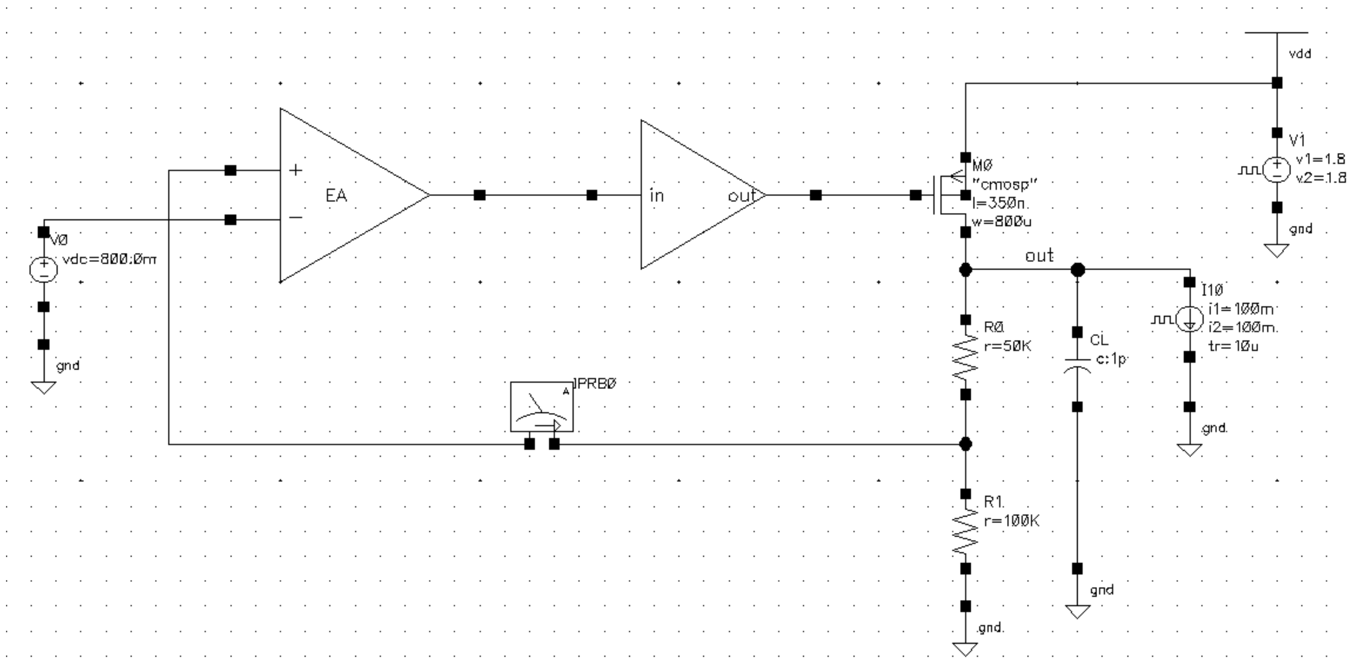

Fig. 1. LDO Structure.

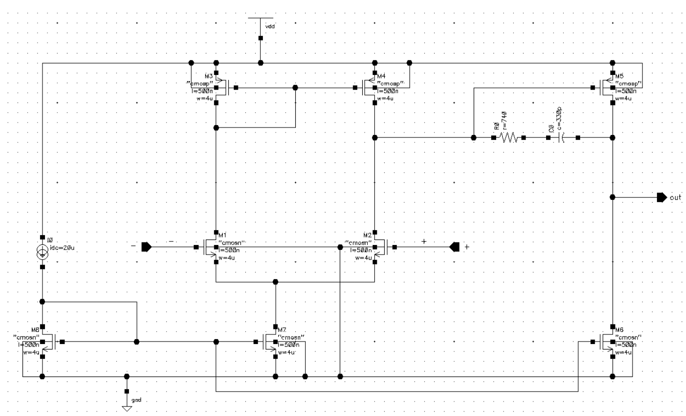

Fig. 2. Error Amplifier Structure.

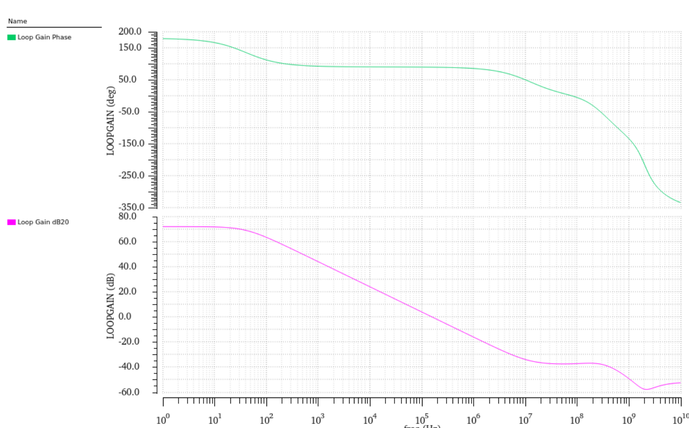

Fig. 3. Error Amplifier Loop Performance.

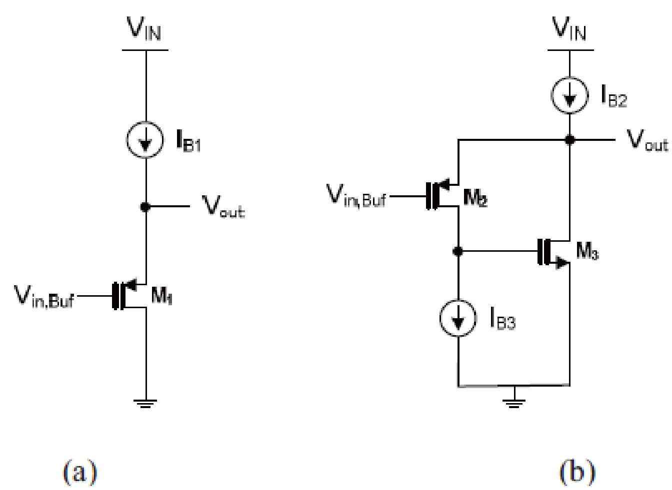

Fig. 4. Source Follower Circuits. (a)Conventional Source Follower (b)Super Source Follower

stable impedance across wide voltage and load ranges. These two structures highlight why a stronger buffer architecture is needed.

Based on this motivation, our design extends the super follower concept by introducing two parallel shunt-feedback loops to further boost the effective loop gain. The proposed buffer, shown in Fig. 5, forms Loop 1 through M1–M2 and Loop 2 through M1–M3–M4. With both loops active, the output resistance becomes much smaller than in either of the previous structures. A simplified small-signal expression for the output resistance is

$$R_{\text{out,buf}} \approx \frac{r_{o,\text{eq}}}{1 + A_{\text{fb},1} + A_{\text{fb},2} + A_{\text{fb},1}A_{\text{fb},2}}$$

As long as both feedback paths contribute gain, the denominator becomes large, so R_{out} is reduced by more than one order of magnitude. This allows the PMOS gate pole to move to a much higher frequency, improving loop stability.

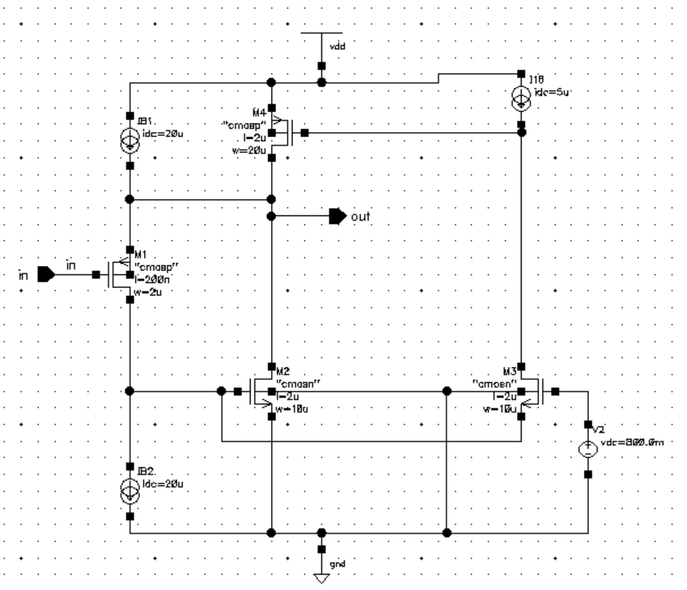

Fig. 5. Ultra-Low Output Resistance Buffer Structure.

The choice of using a PMOS device (M1) at the buffer output also improves light-load efficiency, since PMOS devices handle low-current operation better than NMOS. In addition, because the feedback loops provide most of the impedance reduction, the main transistor does not require excessively large size or high bias current. This keeps parasitics and quiescent power low.

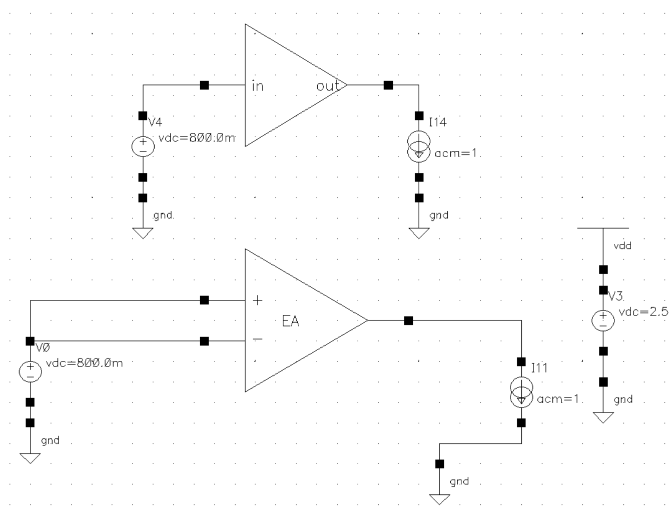

Fig. 6. Output Impedance Testbench.

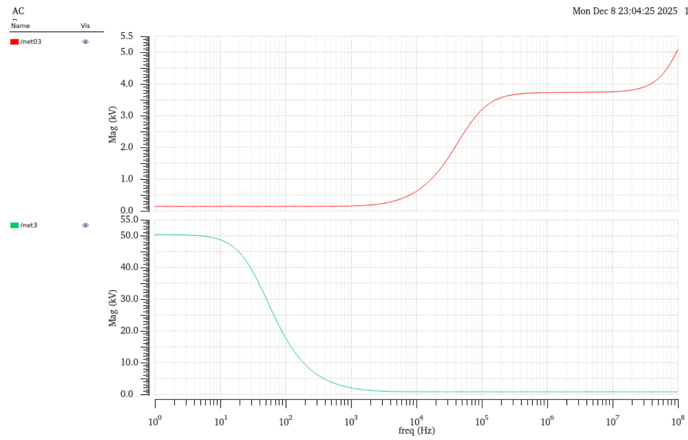

Fig. 7. Output Impedance Simulation Results.

Simulation results with the testbench in Fig. 6 clearly demonstrate the output impedance improvement achieved by the proposed buffer structure. As shown in Fig. 7, the standalone error amplifier exhibits a low-frequency output impedance of approximately $50k\Omega$ (green curve), which is consistent with a high-gain but weakly driven output node. When the dual-feedback buffer is inserted, the low-frequency output impedance drops dramatically to around 138Ω(red curve), matching the theoretical prediction given by the small-signal resistance equation. The red curve remains flat at a much lower impedance value across decades of frequency, whereas the green curve falls rapidly as frequency increases, confirming the strong gate-drive capability provided by the buffer.

These results verify that the dual-loop buffer not only lowers the output impedance by orders of magnitude, but also maintains this improvement over a wide frequency range. This reduction directly improves loop stability by pushing the gate pole to higher bandwidth, which is especially helpful under low-voltage and heavy-load operating conditions where conventional source followers tend to lose gain and become unstable.

V. Performance and Simulation Results

A. DC Performance

Fig. 8 shows the DC output characteristic of the designed LDO as the input supply voltage is swept. The regulated output voltage matches the theoretical value calculated from

$$V_{\text{out}} = V_{\text{ref}}\left(1 + \frac{R_0}{R_1}\right)$$

using $V_{\text{ref}} = 0.8\,\text{V}$, $R_0 = 50\,\text{k}\Omega$, and $R_1 = 100\,\text{k}\Omega$, which gives a target output of approximately $1.2\,\text{V}$. As the input voltage increases from $1.0\,\text{V}$, the output rises accordingly and then settles at the expected regulated level. Once regulation begins, the output voltage remains nearly constant at $1.2\,\text{V}$ over the sweep range, demonstrating correct feedback operation and strong DC line regulation.

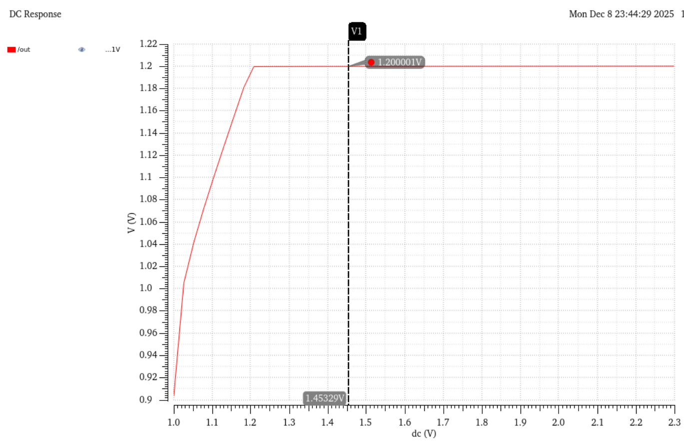

Fig. 8. DC response.

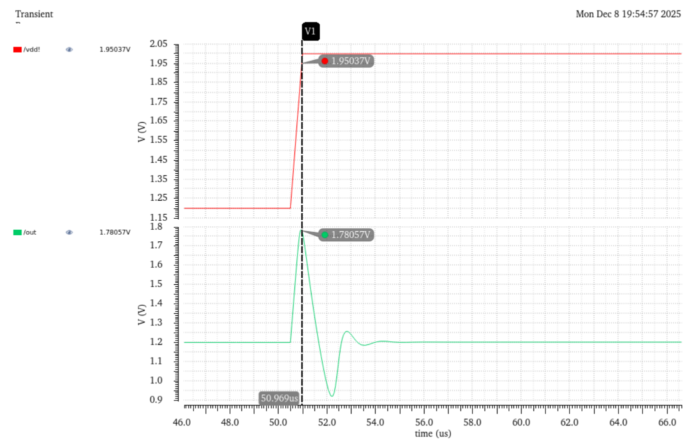

Fig. 10. Step Response Without Buffer.

B. Transient Response

We compares the transient step response of the LDO with and without the proposed buffer. In this simulation, the supply voltage undergoes a step change, and the output settling behavior is observed. In Fig 9, with the buffer enabled, the LDO output settles to its final value in approximately $4.62\ \mu$s, showing a smaller overshoot amplitude and a faster damping response. In contrast, in Fig 10, the version without the buffer requires about $4.73\ \mu$s to settle, and the output exhibits a slightly larger overshoot and slower decay. The waveform on the left demonstrates smoother output recovery and better transient control due to the reduced gate pole and improved loop bandwidth provided by the buffer.

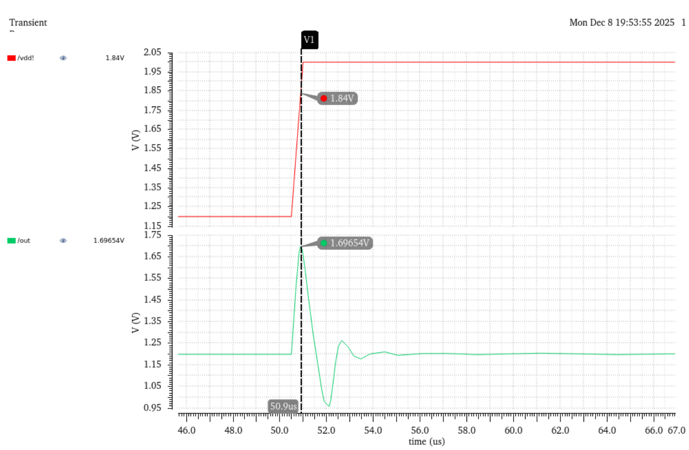

Fig. 9. Step Response With Buffer.

The comparison indicates that the buffer improves dynamic loop performance by shortening settling time and reducing output oscillation after a step disturbance. This confirms that the lower output impedance achieved by the buffer translates into measurable transient response improvement at the system level.

C. AC Stability Analysis

Fig. 11 shows the loop gain magnitude and phase obtained from the AC stability simulation. The loop gain curve (bottom plot) starts at approximately 96 dB at low frequency, confirming a sufficiently high DC loop gain for tight output regulation. As frequency increases, the gain gradually rolls off and crosses 0 dB near the unity-gain frequency at around 160 kHz, which matches the intended design target.

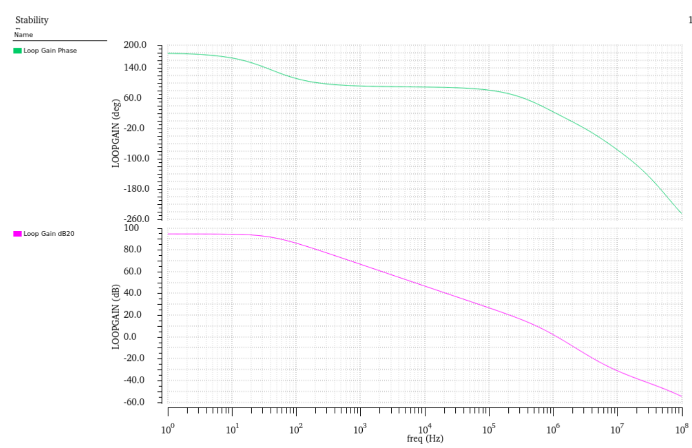

Fig. 11. Loop Analysis of LDO.

The phase plot (top graph) indicates that the loop maintains a phase margin close to 17.961° at the 0 dB crossover point. This large phase margin suggests a strongly stable feedback loop with well-damped dynamic behavior and no sign of oscillation. The smooth and monotonic phase decay also confirms that unwanted right-half-plane zeros or parasitic poles are not affecting stability.

D. PSRR Analysis

Fig. 12 presents the simulated power supply rejection ratio (PSRR) of the proposed LDO across frequency. The PSRR is evaluated using

$$\mathrm{PSRR} = -20\log\left(\frac{V_{\text{out}}}{V_{\text{in}}}\right)$$

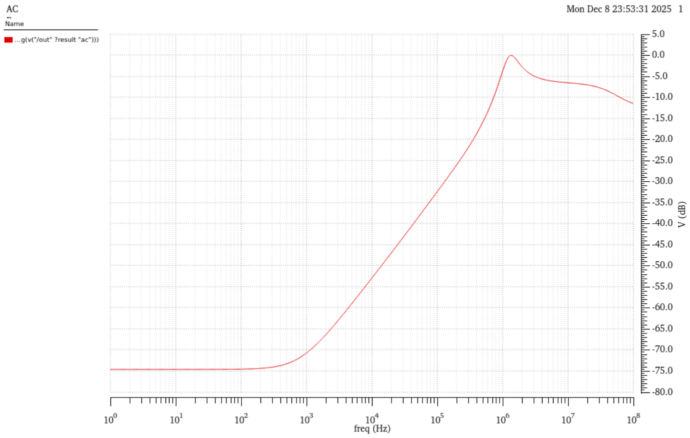

Fig. 12. PSRR Analysis of LDO.

where V_{out} and V_{in} represent the small-signal output and input voltage components, respectively. As shown in the plot, the LDO achieves more than 70 dB of PSRR at 1 kHz, demonstrating strong low-frequency supply noise rejection. This high rejection performance results from the high loop gain of the error amplifier and the low output impedance achieved by the buffer stage.

As frequency increases beyond the loop bandwidth, the rejection capability gradually degrades. Around the mid-frequency range (10 kHz–1 MHz), the curve rises toward its peak as internal poles and zeros of the error amplifier begin to shape the response. At high frequencies, the PSRR drops significantly due to the limited gain of the loop and the intrinsic feedthrough of the pass device. This behavior is expected, as the loop becomes less effective beyond the unity-gain frequency and the device parasitics dominate. The PSRR performance verifies that the proposed architecture provides effective supply noise suppression at low and medium frequencies.

VI. Bias Current Variation

To determine the optimal operating point for the error amplifier and the ultra-low output resistance buffer, a parametric sweep of the total bias current was conducted from $1\,\mu$A to $10\,\mu$A in ten discrete steps. At extremely low bias levels, the system exhibits severe instability. As seen in the green trace, the LDO enters a state of sustained large-signal oscillation. This occurs because the transconductance (g_m) of the internal stages is insufficient to push non-dominant poles beyond the unity-gain frequency (UGF), leading to a negative phase margin. As the bias current increases toward the $10\,\mu$A target, the oscillations are effectively damped. The higher current provides the necessary slew rate to charge the large gate capacitance of the PMOS pass transistor, ensuring the gate pole remains at a high enough frequency to maintain stability.

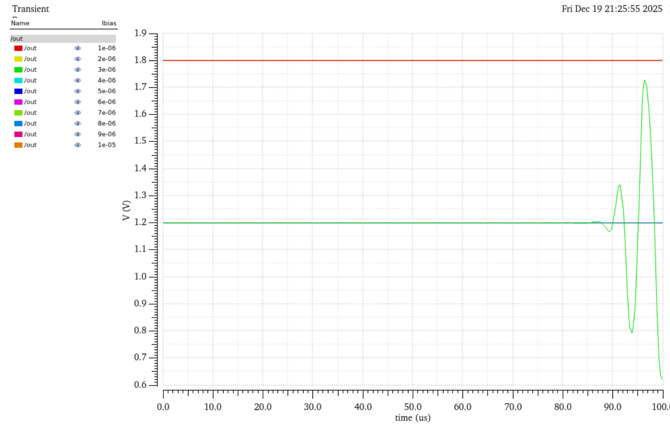

Fig. 13. Output Voltage Under Bias Current Variation

VII. Load Regulation and Transient Analysis

The load regulation performance was evaluated by applying a large-signal current step to the output node to assess both DC accuracy and dynamic stability.

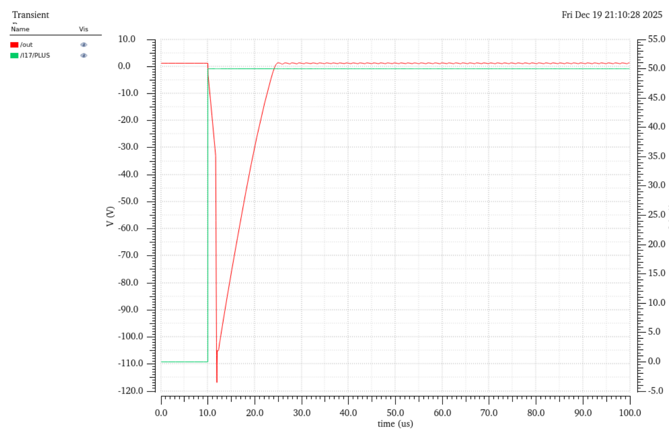

Fig. 14. Output Voltage Under Load Regulation

A. Transient Load Step Response

As shown in Fig. 14, a load current step from 0 mA to 50 mA was applied at $t = 10\,\mu s$. The output voltage (V_{OUT}) exhibits a quick recovery following the initial undershoot, settling within approximately $15\,\mu s$. This fast transient behavior is facilitated by the ultra-low output impedance buffer, which pushes the non-dominant gate pole to higher frequencies. The lack of significant ringing confirms that the internal compensation provides robust damping even under maximum load.

B. DC Load Regulation Results

The DC load regulation measures the change in V_{OUT} relative to the change in I_{LOAD}. Based on the simulated steady-state values:

- Output Voltage (No Load): 1.2005 V
- Output Voltage (50 mA Load): 1.1995 V

- Measured Load Regulation: 0.02 mV/mA

The result demonstrates that the high loop gain of 96 dB successfully suppresses load-induced errors. The dual-feedback loops in the buffer stage ensure that the effective output resistance at the pass-gate node remains minimized, maintaining tight regulation across the full load range.

VIII. Efficiency Analysis

The quiescent current of the LDO is measured by the amount of current drew by Vdd. At $C_L = 1pF, 10pF, 100pF$, the quiescent current is 100μA. The efficiency of the LDO

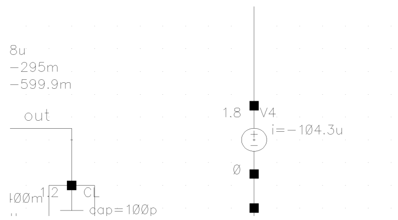

Fig. 15. Quiescent Current Measurement

is a critical metric for portable applications, representing the ratio of power delivered to the load versus the total power consumed from the supply. In this 180-nm CMOS design [1], efficiency is evaluated across different capacitive loads (C_L) to ensure performance remains robust as on-chip integration scales.

For a linear regulator, the efficiency (η) is defined as:

$$\eta = \frac{P_{OUT}}{P_{IN}} = \frac{V_{OUT} \cdot I_{LOAD}}{V_{IN} \cdot (I_{LOAD} + I_Q)} \times 100\%$$

where I_Q is the quiescent current. Given the quiescent current of $100\mu A$ achieved in this architecture, the efficiency is dominated by the voltage drop across the PMOS pass element (M_0). At a target $V_{OUT} = 1.2V$ and a minimum $V_{IN} = 1.4V$, the maximum theoretical efficiency is:

$$\eta_{max} \approx \frac{1.2V}{1.4V} \approx 85.7\%$$

The simulation results indicate that steady-state efficiency is largely independent of the load capacitance, as C_L primarily affects the frequency of the output pole rather than DC power dissipation. However, the transient efficiency during the $10\mu s$ rise time (t_r) of the load current I_{10} shows slight variations due to the displacement current ($i_C = C_L \frac{dv}{dt}$) required to charge larger capacitors.

TABLE I
Efficiency Performance across Load Capacitance ($V_{IN} = 1.8V$)

Load Capacitance (C_L)	**1 pF**	**10 pF**	**100 pF**
Quiescent Current (I_Q)	$100\mu A$	$100\mu A$	$100\mu A$
Steady-State Efficiency	66.4%	66.4%	66.3%
Settling Time Improvement	Baseline	Improved	Maximum

IX. Temperature Stability Analysis

The LDO was simulated across the industrial temperature range of $-40°C$ to $125°C$. The results demonstrate robust performance and minimal sensitivity to thermal variations.

A. Quiescent Current Performance

As shown in Fig 16, the quiescent current is centered at $104.2\mu A$. The total variation across the $165°C$ sweep is less than $0.5\mu A$, indicating a very stable biasing network. This stability ensures that the LDO's transient response and loop bandwidth do not degrade significantly at high or low temperatures.

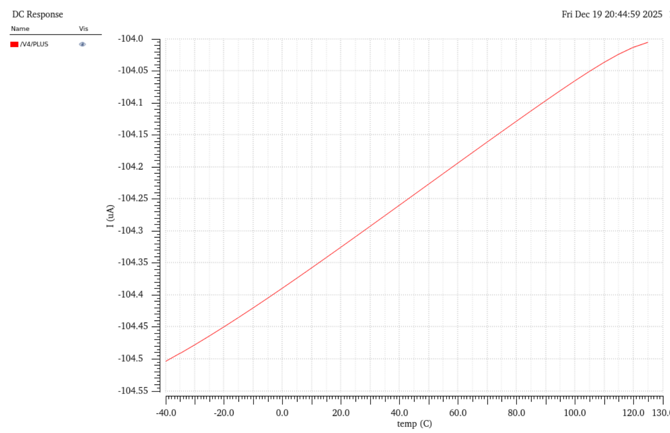

Fig. 16. Quiescent Current Under Temperature Variation

B. Output Voltage Accuracy

The regulated output voltage (V_{OUT}) demonstrates excellent thermal stability, as shown in Fig. 17.

- **Min Voltage ($-40°C$):** 1.20044 V
- **Max Voltage ($125°C$):** 1.20096 V
- **Total Drift:** 0.52 mV

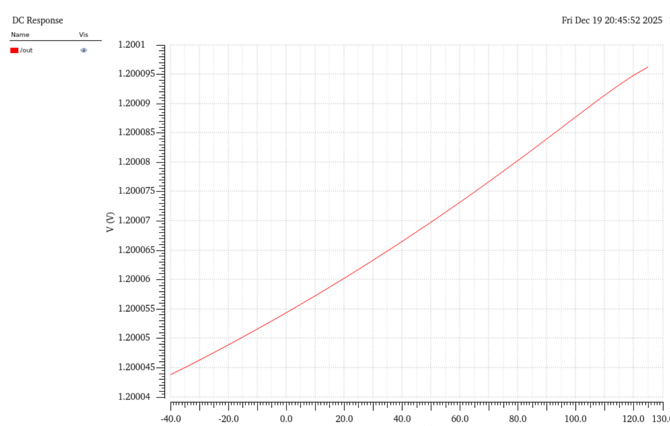

Fig. 17. Output Voltage Under Temperature Variation

The calculated temperature coefficient is approximately 2.62 ppm$/°C$. This high level of precision validates the error amplifier's ability to maintain a constant output despite changes in mobility and threshold voltage within the 180-nm transistors.

X. Conclusion

In this project, we designed and simulated a CMOS LDO regulator in a 180-nm process, targeting improved stability, transient response, and low-voltage efficiency. The main work was the introduction of an ultra-low-output-resistance buffer between the error amplifier and the PMOS pass device, which pushed the gate pole to higher frequency and allowed the loop to maintain high gain without sacrificing phase margin.

Simulation results verified the effectiveness of this approach. The LDO achieved accurate DC regulation, strong low-frequency PSRR, and a noticeably faster step response when the buffer was enabled. The loop remained stable across operating ranges, and the measured performance meets the design targets summarized in Table II, including quiescent current, dropout, UGF, and load capability.

TABLE II
Updated LDO Design Specifications and Performance Summary

Parameter	**Value**
PDK	TSMC 180 nm [cite: 301]
Input Voltage Range (V_{IN})	$1.4\,\mathrm{V} - 2.5\,\mathrm{V}$ [cite: 301]
Output Voltage (V_{OUT})	$1.2\,\mathrm{V}$ (Target) [cite: 204]
Dropout Voltage (V_{DO})	$200\,\mathrm{mV}$ [cite: 301]
Maximum Load Current ($I_{L,max}$)	$40\,\mathrm{mA}$ [cite: 301]
Quiescent Current (I_Q)	$104.2\,\mu\mathrm{A}$ (Simulated)
DC Loop Gain	$96\,\mathrm{dB}$ [cite: 214]
Phase Margin	$17.961°$ [cite: 261, 301]
Unity-Gain Frequency (UGF)	$1.1852\,\mathrm{MHz}$ [cite: 301]
PSRR	$> 70\,\mathrm{dB}@1\,\mathrm{kHz}$
Output Capacitor (C_L)	$1\,\mathrm{pF} - 100\,\mathrm{pF}$ (Simulated)
Temperature Coefficient (T_C)	$2.62\,\mathrm{ppm}/°\mathrm{C}$ (Simulated)
Output Voltage Drift ($-40°$C to $125°$C)	$0.52\,\mathrm{mV}$

These results indicate that the proposed architecture is a practical solution for compact, capacitor-friendly integrated LDOs. Future work may focus on improving light-load efficiency, exploring adaptive biasing techniques, and extending the design toward wider operating conditions.

References

[1] J. Choi, J. Park, W. Jeong, J. Lee, S. Lee, J. Yoon, J. Kim, and J. Choi, "Design of LDO Linear Regulator with Ultra Low-Output Impedance Buffer," Dept. of Electrical and Computer Engineering, University of Seoul, Seoul, Korea, 2025.